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From situational awareness to systemic stability: A phased institutional pathway for space traffic management

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ABSTRACT

Rapid growth in satellite deployments has transformed Earth orbit from a sparse operational environment into a continuously negotiated shared domain. Modern tracking networks increasingly detect conjunction risks, yet collision avoidance remains dependent on bilateral coordination between operators rather than predictable procedural rules. Space situational awareness (SSA) is therefore indispensable but incomplete: it provides probabilistic knowledge of orbital conditions but does not yet constitute a traffic management regime capable of ensuring consistent, system-level predictability.

This paper examines how space traffic management (STM) can be developed by converting awareness into structured coordination through phased institutional design rather than immediate supranational authority. It distinguishes SSA from STM and identifies short-term improvements that establish interpretive interoperability without redistributing authority.

To address scaling constraints, the paper introduces a cooperative Orbital Stability Assessment (OSA) producing a periodic Orbital Stability Index (OSI). As a pre-institutional diagnostic pilot involving civil safety service providers, commercial SSA analytics providers, and spacecraft operators, OSA would evaluate system-level coordination conditions by measuring coordination burden rather than incident occurrence, while OSI would synthesize those observations into a periodic assessment of whether coordination remains predictable and scalable as operational demands increase. The framework is diagnostic rather than regulatory and is intended to provide a shared empirical basis for subsequent governance and institutional decision-making.

Building on the recommendations of the International Institute of Space Law (IISL) Working Group on the Legal Aspects of Artificial Intelligence in Space, the paper evaluates governance architectures and proposes a phased pathway toward a Space Traffic Management Authority (STMA). STMA is conceived as a centralized standard-setting institution without real-time operational control, supported by distributed implementation, market-based compliance incentives, and interoperability mechanisms. Drawing on the procedural model of the International Civil Aviation Organization (ICAO), it would define global operational expectations while implementation remains national and does not require immediate treaty formation.

The governance pathway is explicitly phased: interoperability through Global Navigation Satellite Systems (GNSS)-type reference layers; convergence through processes analogous to the Joint Authorities for Rulemaking on Unmanned Systems (JARUS) and standards bodies such as the International Organization for Standardization (ISO); followed by the formation of STMA as a centralized institutional standard-setting body; then hybrid compliance reinforced through distributed regulatory and market mechanisms; and, once operational practices stabilize and verification mechanisms mature, selective consolidation analogous to the International Maritime Organization (IMO). The paper further examines implementation challenges associated with participation incentives, insurance-based compliance mechanisms, free-rider behavior, and geopolitical fragmentation, arguing that institutional success should be evaluated not solely by universal participation, but by the ability to facilitate predictable coordination across diverse governance architectures.

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1. Introduction

Earth orbit has entered a phase in which operational safety depends less on the detection of hazards than on the anticipation of other actors' behavior. Satellite operators now routinely receive conjunction warnings derived from multiple tracking networks, yet collision avoidance decisions continue to be resolved through direct communication rather than through predefined procedural hierarchy [1]. The orbital domain therefore exhibits a degree of situational awareness without institutionalized traffic management. In this context, "management" refers to the structuring of coordination through shared expectations rather than centralized operational control.

Early formulations of space traffic management (STM) emphasized that surveillance alone is insufficient to ensure coordinated safety outcomes. The 2006 IAA Cosmic Study defined STM as a set of technical and regulatory provisions intended to promote safe access to, operations in, and return from outer space free from physical and radio-frequency interference, thereby framing STM as a governance challenge extending beyond the tracking of objects alone [2]. Building on this foundation, the 2018 IAA roadmap examined pathways for STM implementation, emphasizing the importance of interoperability, information-sharing, technical standards, and institutional development, while recognizing that traffic management capabilities would likely emerge progressively through the convergence of operational practice, technical coordination, and regulatory measures [3]. Together, these studies frame STM not merely as a problem of sensing, but as one of coordination, governance, and institutional evolution.

Modern space situational awareness (SSA) capabilities increasingly identify potential collisions, but they remain subject to limitations in coverage, data quality, and transparency. Even where detection is sufficient, SSA does not ensure consistent interpretation of risk or predictable allocation of maneuver responsibility [1,4,5]. Operators continue to resolve high-risk encounters on a case-by-case basis, often under compressed timelines and with incomplete confidence in counterpart decisions [1,4,5]. Safety performance therefore depends more on cooperative practice than on structured governance.

That distinction becomes increasingly consequential as orbital congestion grows. When the number of active objects was comparatively low, ad hoc coordination scaled with acceptable reliability. In a denser environment, however, repeated negotiation introduces systemic fragility even where individual encounters are successfully resolved. The challenge is no longer limited to preventing isolated incidents, but to sustaining predictable behavior across continuous interactions. As conjunction frequency increases, this challenge is further intensified by the growing role of automated and AI-assisted decision-support systems, which may accelerate response while also amplifying the consequences of divergent assumptions, opaque models, and inconsistent thresholds for intervention.

International space law establishes shared responsibility for space activities but does not prescribe specific operational coordination procedures: the Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space requires due regard and avoidance of harmful interference, while the Convention on Registration of Objects Launched into Outer Space establishes identification and transparency mechanisms [6,7]. More recently, the Guidelines for the Long-Term Sustainability of Outer Space Activities adopted through the United Nations Committee on the Peaceful Uses of Outer Space encourage information exchange, conjunction assessment, and contactability among operators [8]. These instruments collectively define standards of conduct but not mechanisms of execution. They presuppose awareness, yet they do not institutionalize traffic management.

Operational practice has therefore partially filled this gap, though in a fragmented and non-uniform manner. Civil safety services such as the United States Traffic Coordination System for Space (TraCSS) and the European Union Space Surveillance and Tracking (EU SST) system distribute conjunction warnings, while commercial providers refine predictions through proprietary analytics. Operators coordinate maneuvers directly with their counterparts, frequently confirming intent through bilateral communication [1,4,5]. The system functions, but its reliability derives primarily from professional norms rather than structured procedure; safety emerges from cooperation rather than from shared rules [1,4,5].

The rapid expansion of large-scale satellite constellations, including emerging mega-constellations, places this equilibrium under strain [9]. Increasing numbers of spacecraft, combined with growing levels of onboard autonomy, are increasing encounter frequency, compressing response timelines, and magnifying the effects of divergent interpretations of risk [10,11]. As orbital operations increasingly resemble continuous maneuver management rather than episodic coordination, predictability becomes more important than precision alone. A technically accurate warning that produces incompatible operational responses cannot ensure safety at scale.

The question, therefore, is not whether coordination exists. Functional coordination already occurs. The challenge is to transform negotiated coordination into a stable governance framework capable of scaling with orbital density. This transformation is unlikely to occur through the creation of a supranational operational authority; States retain jurisdiction over national space activities and are unlikely to transfer real-time spacecraft control [12]. Instead, institutional development must proceed through a structured progression: shared technical interpretation, procedural harmonization, and progressively formalized standard-setting before selective legal consolidation.

To support this transition, this paper introduces a cooperative Orbital Stability Assessment (OSA) as a pre-institutional diagnostic framework for evaluating system-level coordination conditions, producing a periodic Orbital Stability Index (OSI) that synthesizes relevant indicators into an assessment of whether coordination remains predictable and scalable as operational demands increase. Building on this empirical foundation, and on the recommendations of the International Institute of Space Law Working Group on the Legal Aspects of Artificial Intelligence in Space Activities, the paper advances a phased pathway toward a Space Traffic Management Authority (STMA), understood not as a centralized traffic controller but as an institutional locus for coordinated rulemaking [13].

In that respect, the proposed architecture is closer to the standard-setting role performed by the International Civil Aviation Organization (ICAO) than to operational command: centralized development of common standards, transparency expectations, and oversight mechanisms, with implementation remaining anchored in national law.

The challenge is not only to detect risk, but to ensure that it is interpreted and acted upon consistently across operators. As orbital activity increases, coordination without shared expectations becomes progressively less reliable. This paper therefore develops a phased institutional pathway to stabilize coordination and enable it to scale.

Importantly, the viability of such a framework cannot be evaluated solely according to its ability to achieve universal participation. Space activities increasingly intersect with national security, industrial policy, economic competitiveness, and strategic autonomy, raising the possibility that multiple governance architectures may emerge rather than a single globally integrated regime. A durable STM framework must therefore be capable not only of supporting coordination among participating actors, but also of facilitating predictable interactions across institutional boundaries where participation remains partial. The challenge is not simply institutional formation, but institutional resilience under conditions of fragmentation.

2. Conceptual distinction between SSA and STM

The terms SSA and STM are frequently used interchangeably in policy discussions, yet they describe fundamentally different functions. The distinction is not semantic. Conflating awareness with management can create the impression that a traffic management regime already

exists, whereas the current framework remains informational rather than procedural.

SSA concerns knowledge of the orbital environment. It encompasses detection, tracking, cataloging, and prediction of object motion [1,4,5]. Modern SSA integrates radar and optical observations, orbit determination models, and data fusion processes to estimate position and uncertainty over time [1,4,5]. Public safety services distribute this information through conjunction data messages and related analytical products, enabling operators to evaluate collision probability and determine whether mitigation is necessary.

STM, by contrast, concerns the broader framework through which space activities are coordinated and governed to support safe and sustainable operations. Work undertaken through the IAF-IAA-IISL Space Traffic Management Technical Committee (TC26) distinguishes STM from related functions including SSA and space traffic coordination (STC), defining STM as an assurance value chain composed of STC and regulation and licensing functions and dependent upon continuous SSA [14]. TC26 terminology also distinguishes space environment preservation (SEP), encompassing debris mitigation and remediation, as an activity directed toward preserving and sustaining the space operations environment [14]. The subsequent TC26 synthesis reflects the broader initiative through which these and other technical, operational, regulatory, and sustainability dimensions of STM were examined [15].

Consistent with these distinctions, this paper examines STM as an institutional and operational framework rather than equating it with SSA or bilateral operator coordination. Effective STM requires predictable procedures and shared operational reference points, including agreed approaches to risk interpretation, standardized expectations for coordination, and consistent practices concerning maneuver responsibility, notification timelines, and response actions [13]. The analysis developed in the following sections therefore considers how existing SSA capabilities and coordination practices can support progressively more structured forms of STM governance. Awareness informs decisions; coordination structures interaction; STM provides the broader framework through which those functions can be integrated and governed.

The present orbital safety framework operates primarily within the first category. Civil services such as the United States Traffic Coordination System for Space (TraCSS) and the European Union Space Surveillance and Tracking (EU SST) system provide conjunction data to a broad range of operators, while commercial analytics providers refine predictions using alternative modeling assumptions [1]. None of these systems, however, determines maneuver responsibility in practice. Operational decisions remain with spacecraft operators subject to national authorization regimes.

This separation reflects the legal structure of space activities. Under Article VI of the Outer Space Treaty, states bear international responsibility for national space activities and must authorize and continually supervise them [6]. Maneuver decisions therefore remain under operator control within domestic licensing frameworks rather than international direction. Existing safety services function as advisory mechanisms rather than operational authorities.

The difference becomes evident during close approaches. Conjunction warnings initiate coordination but do not resolve it. Operators compare internal assessments, clarify intent, and negotiate responsibility for maneuvering. Identical data may produce divergent operational conclusions, particularly where mission constraints or maneuver capability differ [16]. Safety is often achieved through agreement rather than predefined hierarchy [17].

Historically, this arrangement generally proved adequate. Sparse traffic allowed time for negotiation, and professional norms supported cooperative behavior [18]. Increasing object density alters that condition. Encounter frequency rises, response timelines compress, and interactions become continuous rather than occasional [16]. Under these conditions, awareness alone does not scale into management.

International guidance recognizes the importance of coordination without prescribing operational mechanisms. The Long-Term Sustainability Guidelines encourage information exchange, communication among operators, and conjunction assessment practices [8]. They are premised on the ability of actors to translate shared information into compatible conduct. As encounter frequency increases, that assumption becomes more difficult to sustain. Reliable safety requires not only knowledge of other objects but shared expectations of response.

The governance challenge therefore lies not in improving detection capability but in converting informational awareness into predictable operational conduct. The next section examines how current operational practice reveals the limits of awareness-based safety despite increasingly capable tracking systems.

3. From situational awareness to traffic management: the operational and governance gap

Having distinguished situational awareness from traffic management in conceptual terms, the practical question is how the current system functions operationally. The orbital environment demonstrates that awareness capabilities have advanced substantially, yet predictable coordination has not emerged automatically from improved detection. The resulting condition is neither an absence of safety nor the presence of a traffic management regime, but an intermediate state in which operational reliability depends on human coordination layered on technical outputs.

The following analysis therefore shifts from definition to operation and examines the governance gap through four connected dimensions: the capabilities and remaining operational limitations of existing SSA systems (§3.1); the resulting gap between improved awareness and predictable coordination (§3.2); the international legal and policy instruments that shape existing expectations of conduct without establishing an operational STM framework (§3.3); and the additional coordination pressures created by increasing reliance on automated and AI-assisted decision-making (§3.4). Taken together, the analysis of these dimensions explains why increasingly capable situational awareness does not, by itself, produce scalable traffic management and establishes the operational and institutional problem that the measures proposed in Section 4 are intended to address.

3.1. *Where SSA succeeds, and where it remains operationally fragile*

Over the past decade, SSA capability has expanded significantly across governmental and commercial sectors. The most consequential development has been the emergence of civil safety services designed to support non-military operations at scale. Certain public SSA providers now offer conjunction-related services to a broad range of operators without direct cost while also catalyzing commercial SSA markets [1].

The existence of these systems, however, does not eliminate operational fragility. Three categories of constraint persist even among sophisticated operators.

First, precision and timeliness remain uneven. Decision-grade conjunction analysis depends on the quality of state vectors, realistic covariance characterization, and rapid updates from owner-operator ephemerides. The joint technical comparison of EU SST and TraCSS identifies operator-provided ephemerides and covariance realism as ongoing areas for improvement rather than settled capabilities [1]. Divergent assumptions about uncertainty propagation can materially affect predicted miss distances and collision probabilities. A conjunction event may appear urgent under one modeling baseline and tolerable under another. This variability does not reflect deficiencies in analysis; it reflects incomplete interoperability in uncertainty modeling.

Second, coordination pathways remain structurally bilateral. Public safety services distribute conjunction data messages and screening outputs, but resolution of conjunctions between active spacecraft frequently depends on direct operator communication. The joint technical comparison of EU SST and TraCSS documents differences in

contact information sharing, maneuver screening support, and candidate maneuver analysis between the two systems, demonstrating convergence in direction but not uniformity in implementation [1]. In practice, high-risk events often require rapid human coordination layered atop automated screening systems. The technical detection of risk does not eliminate the need for negotiated response.

Third, strategic and security constraints shape data sharing. Civil SSA infrastructures in several jurisdictions remain dependent on military-origin tracking inputs [1]. Certain government missions or commercially sensitive operators may limit ephemeris disclosure or maneuver transparency. This reality must be treated as a structural design parameter rather than a normative failure. The safety system therefore operates with asymmetric transparency: some actors provide high-fidelity updates; others provide limited public detail. As orbital density increases, asymmetric transparency amplifies systemic uncertainty.

Recent private initiatives illustrate both the benefits and limits of expanded SSA access. For example, SpaceX has introduced a space situational awareness system ("Stargaze") that generates conjunction data messages and supports a traffic coordination platform through the sharing of screening outputs with participating operators [19]. Such initiatives represent a meaningful advancement in data availability and responsiveness, with the potential to enhance visibility into conjunction risk and lower barriers for emerging actors. However, a single private system cannot establish cross-jurisdictional standards, resolve conflicts of interest between competitors, or define neutral interoperability requirements. Parallel analytic baselines may improve tooling while simultaneously reinforcing fragmentation if interpretive standards diverge. Improved visibility does not by itself generate governance coherence.

3.2. The governance gap between awareness and management

The relationship between SSA and STM can therefore be summarized as follows: SSA produces probabilistic knowledge; STM governs how that knowledge is translated into structured behavior. The current system exhibits a governance gap because shared warnings do not guarantee shared responses.

This governance gap has also been reflected in multilateral discussions. A 2015 working paper submitted to the Legal Subcommittee of the United Nations Committee on the Peaceful Uses of Outer Space informed discussions that led to the establishment of a dedicated agenda item on STM in 2016, providing a forum for continued exchange of views on coordination concepts, operational practices, and emerging policy approaches [20]. More recently, multilateral coordination efforts have continued to evolve. At the 63rd session of the Scientific and Technical Subcommittee of COPUOS, the United Arab Emirates chaired an international Expert Group on Space Situational Awareness, bringing together states and organizations to advance data-sharing practices and coordination mechanisms relevant to orbital safety [21].

Two structural features intensify this gap. First, orbital safety is inherently interdependent: one actor's opacity or delayed response can elevate risk across multiple networks, including those operating under robust internal compliance frameworks. Second, the orbital environment lacks natural chokepoints. Unlike aviation, which relies on sovereign airspace boundaries, or maritime systems anchored to port state control, orbital motion is continuous and geography-independent. There are no in-orbit checkpoints at which compliance can be physically enforced. As a result, enforcement levers attach primarily to Earth-side dependencies, including launch authorization, spectrum allocation, insurance markets, financing conditions, and ground infrastructure access.

For these reasons, STM must be evaluated as institutional design rather than sensor architecture. The problem is not insufficient detection; it is insufficient predictability in how risk assessments are operationalized. The remainder of this paper therefore treats SSA as the operational substrate and examines how governance mechanisms can convert fragmented coordination into scalable, system-level stability.

3.3. Baseline international instruments influencing SSA and STM

Although STM is often framed as a future regime, existing international instruments already shape expectations of conduct.

At the treaty level, the foundational obligations remain those established under the Outer Space Treaty, including state responsibility for national activities (Article VI) and the obligation to conduct activities with due regard to the corresponding interests of all other States Parties (Article IX) [6]. The Registration Convention reinforces transparency by requiring states to furnish identifying information about launched objects [7]. These instruments, among the most relevant in this context, do not prescribe operational coordination procedures, but they establish the normative substrate against which responsible behavior is assessed.

At the guideline level, the Space Debris Mitigation Guidelines adopted by the United Nations Committee on the Peaceful Uses of Outer Space provide a globally recognized baseline for debris prevention, post-mission disposal, and avoidance of intentional debris-generating events [22]. These guidelines are non-binding and are intended to be implemented through national mechanisms to the greatest extent practicable. Their influence is therefore mediated through licensing, procurement standards, and regulatory practice rather than direct enforcement. The Long-Term Sustainability (LTS) Guidelines further encourage information exchange, communication among operators, and conjunction assessment practices [8].

At the regional level, emerging regulatory initiatives reflect a shift toward more structured operational requirements. The European Commission's proposed Space Act introduces provisions related to collision avoidance, data sharing, and operational safety at the level of individual operators, indicating a move toward greater standardization within a defined regulatory framework without constituting a comprehensive traffic management regime [23].

Collectively, these instruments create a vocabulary of responsibility and sustainability. They do not, however, institutionalize interoperable maneuver coordination, shared risk thresholds, or harmonized service definitions. They provide a normative foundation for SSA and debris mitigation; they do not constitute STM governance.

The limitations of the existing framework become more consequential as conjunction assessment and maneuver decision-making are increasingly mediated by automated and AI-assisted systems. In the absence of shared operational standards, increasing automation may amplify existing differences in how operators and systems interpret risk and coordinate action, particularly as decision timelines become more compressed. Automation thus intensifies the governance gap identified above by increasing the importance of predictable interpretation, validation, and coordination across heterogeneous systems.

3.4. Automation, autonomy, and limits of human control

The operational limitations of the current SSA framework are further intensified by the increasing reliance on automated and AI-assisted decision-support systems. While SSA provides probabilistic information about conjunction risk, the translation of that information into action is increasingly mediated by systems operating under compressed timelines that may exceed human response capacity.

As conjunction frequency increases, these systems are expected to evaluate risk and generate maneuver recommendations at speeds that limit the ability of operators to independently verify, interpret, or override outputs before action is required [24]. This creates structural constraints on direct human control, particularly in high-tempo environments where delayed response may itself increase risk.

At the same time, reliance on such systems introduces risks of automation bias, whereby human decision-makers defer to algorithmic recommendations even when underlying assumptions, data quality, or uncertainty representations differ [25]. Institutional incentives may

further discourage intervention, as operators face asymmetric risk in overriding automated recommendations under uncertainty [26]. These risks are compounded by the opaque or "black box" nature of advanced models, including machine learning systems whose internal decision pathways may not be fully interpretable even to their designers [25].

In a distributed operational environment, heterogeneous models, proprietary analytics, and differing training data may produce divergent automated responses to identical conjunction scenarios, thereby reducing predictability rather than enhancing it [27,28]. The challenge is therefore not the adoption of automation itself, but the absence of shared frameworks governing how such systems interpret and act upon risk.

Ensuring meaningful human control in this context requires more than the formal presence of human oversight. It depends on institutional mechanisms that define validation standards, transparency expectations, and conditions for human override within a shared operational framework. Analogous challenges have been addressed in other safety-critical domains through structured software-assurance frameworks, such as the DO-178C certification process used in civil aviation, which establishes rigorous requirements for the development, verification, and validation of software performing safety-critical functions. Without such alignment, automation risks amplifying coordination uncertainty rather than resolving it. The following section therefore examines how technical standardization can establish the common interpretive layer necessary to support both human and automated decision-making in a scalable traffic management system.

4. Immediate operational improvements to Space Situational Awareness

The following measures are not presented as isolated technical improvements, but as components of a shared operational understanding necessary for scalable coordination. The analysis above demonstrates that the present safety constraint arises from a combination of factors: limitations in tracking coverage, variability in data quality and timeliness, restrictions on data sharing, and the absence of shared interpretation of SSA outputs. Operators routinely receive technically valid conjunction warnings yet must still negotiate their behavioral implications. The most realistic near-term progress therefore lies neither in creating new global institutions nor in awaiting a comprehensive traffic management treaty, but in stabilizing the practical meaning of existing SSA outputs.

At this stage, the priority is interpretive interoperability: similar conjunction information should produce comparable operational expectations across jurisdictions and operator classes. Without that condition, coordination mechanisms cannot reliably scale into traffic management. The measures described below do not redistribute authority or impose binding obligations. Instead, they translate cooperative principles already reflected in international guidance into operational conventions capable of supporting a coordinated, multi-actor safety framework.

4.1. Conjunction risk categories as shared operational reference

Collision probability presently functions primarily as a technical metric rather than a behavioral signal. Operators may interpret similar probability estimates differently according to internal mission constraints, propulsion margins, and organizational risk tolerance [17]. As a result, two spacecraft may rationally adopt different courses of action while both complying with existing guidance. The resulting unpredictability arises not from insufficient data alone, but from the absence of shared interpretation.

A practical improvement could be to associate statistical outputs with common response semantics. Conjunction messages could incorporate standardized severity categories derived from probability, miss distance, and maneuverability assumptions. The objective is not to prescribe maneuvers or allocate responsibility, but to establish a shared understanding of how conjunction risk should be interpreted across operators. Such categories would indicate whether an encounter is informational, cautionary, or requires coordination.

These classifications would function as a shared behavioral reference. Operators receiving the same alert could better anticipate how others are likely to interpret the event, reducing negotiation time and improving predictability without centralized command authority. Technical standardization bodies concerned with interoperability, including the Consultative Committee for Space Data Systems (CCSDS) and relevant International Organization for Standardization (ISO) processes, offer appropriate venues, as the task concerns message semantics rather than legal obligation [29,30].

4.2. Structured transparency windows for maneuver coordination

Current maneuver communication practices vary significantly. Some operators share ephemerides in advance, others confirm maneuvers after execution, and some disclose information only upon request [1]. These differences reflect legitimate commercial, operational, and security considerations; however, they also complicate conjunction resolution during time-sensitive encounters because silence cannot readily be distinguished from disagreement, delay, or non-responsiveness.

Operational coordination improves when transparency is predictable rather than discretionary. The objective is not to prescribe comprehensive disclosure requirements, but to establish when communication should occur while ensuring that the information exchanged is sufficient and reliable for meaningful coordination. Linking communication expectations to severity categories could allow confidentiality to be preserved while ensuring that coordination windows exist when safety risk increases. Above defined risk thresholds, operators could be expected to communicate maneuver intent within a recommended timeframe and to confirm execution within a comparable post-event interval.

This approach operationalizes the cooperative logic reflected in the Long-Term Sustainability Guidelines by introducing temporal structure into communication and information exchange. It does not prescribe a comprehensive or uniform disclosure standard; rather, it establishes when communication should occur so that the absence of response carries operational meaning, while the minimum information and data-quality requirements necessary for meaningful coordination are addressed separately below.

4.3. Minimum ephemeris quality conventions

Many disagreements in conjunction resolution arise from differing uncertainty assumptions rather than conflicting safety priorities. Public SSA services rely on observational catalogs augmented by operator-provided ephemerides; however, differences in ephemeris quality, update practices, and uncertainty characterization can lead to divergent risk assessments [1,4,5]. Divergent assumptions can produce different risk predictions that operators must reconcile manually.

Improving ephemeris consistency may offer greater safety benefit than expanding sensing coverage alone. Common conventions for uncertainty description, update cadence, and maneuver flagging would enable independent safety services to generate more comparable predictions. The EU SST–TraCSS comparative study identifies ephemeris quality and covariance realism as persistent operational challenges across systems, indicating a shared technical limitation rather than a jurisdiction-specific one [1].

Adopting minimum quality expectations would therefore enhance safety without altering legal responsibilities. The improvement lies in enabling distributed providers to interpret operator data more consistently, allowing a network of services to function more coherently.

4.4. Common terminology across SSA services

Even closely aligned safety providers have required joint technical comparison to reconcile definitions of screening events, emergency thresholds, and alert classifications. Divergent terminology can require operators to translate between systems before determining whether action is required, introducing delay precisely when time sensitivity increases [1].

Establishing a shared lexicon for conjunction messaging would reduce this friction. Common definitions for screening events, coordination-critical encounters, and emergency conditions would allow operators to treat notifications from different services as functionally comparable. Terminology alignment does not unify system architecture; it aligns interpretation. Historically, such convergence often precedes institutional governance because actors can more readily coordinate once signals carry sufficiently consistent meaning.

4.5. The role and limits of private SSA capability

Commercial SSA providers are increasingly expanding data access and analytic capability. These contributions improve visibility and support operators lacking internal expertise. However, private services cannot establish neutral expectations across competing actors, define authoritative thresholds across jurisdictions, or resolve conflicting interpretations of risk.

Expanded data availability therefore reinforces the need for shared conventions. Tools can enhance situational awareness; only collectively defined standards can produce more predictable interaction. A federated SSA architecture remains viable only if its outputs carry common meaning independent of provider.

Collectively, these measures transform SSA from informational awareness into an operational reference layer. They remain voluntary, technically grounded, and compatible with national authority while significantly reducing reliance on ad hoc coordination. Establishing this interpretive baseline enables consistent cooperative behavior rather than incidental coordination. The following section examines how such convergence can be evaluated through a cooperative pilot framework prior to broader institutional adoption.

5. Pilot framework: cooperative Orbital Stability Assessment and index

The preceding sections demonstrated that current space situational awareness practice often enables successful conjunction avoidance but not predictably. Operators routinely prevent collisions, yet outcomes depend on negotiation timing, modeling assumptions, and the willingness of counterpart operators to respond. Civil safety systems distribute warnings, commercial providers reconcile discrepancies, and operators coordinate maneuvers; however, the system functions through repeated case-by-case interaction rather than through a shared operational interpretation of risk. As orbital populations increase, this distinction becomes decisive. A domain may remain largely collision-free while becoming operationally unstable.

The limitation is structural: the operational use of SSA focuses on discrete encounters, whereas long-term sustainability depends on whether coordination scales with increasing density. Present safety indicators therefore measure incident avoidance rather than systemic operability. The relevant policy question is not solely whether spacecraft collide, but whether operators can continue to anticipate one another's behavior within workable decision timelines.

Other transportation domains reflect a comparable transition from incident avoidance toward system-level safety measurement [31,32]. Aviation safety evolved from preventing individual mid-air collisions toward monitoring traffic density and separation performance, and maritime safety progressed from navigational advisories toward internationally recognized safety baselines supported by shared operational measurements prior to legal obligation. In each case, shared observation informed the development of standards, which in turn supported institutional authority. Space operations do not yet possess an equivalent intermediate observational layer.

To bridge the gap between technical harmonization and institutional traffic management, this paper proposes a cooperative Orbital Stability Assessment (OSA) as a pre-institutional diagnostic framework for evaluating system-level coordination conditions within an orbital regime. OSA would use aggregated operational indicators to assess whether the coordination processes through which collision risk is managed remain predictable and scalable as operational demands increase. Its resulting Orbital Stability Index (OSI) would synthesize those observations into a periodic system-level assessment. Neither OSA nor OSI would manage traffic, regulate or evaluate individual operators, assign maneuver responsibility, prescribe operational conduct, or determine compliance. Their function is diagnostic: to provide an empirical basis for understanding when existing coordination arrangements are coming under systemic strain.

5.1. From encounter management to system-level coordination conditions

Current SSA services focus on conjunction screening, probability estimation, fragmentation monitoring, and re-entry tracking. These activities enable operators to address individual encounters but do not measure whether coordination effort itself is increasing as a function of encounter frequency and operator density. Two systems may prevent the same number of collisions while one requires substantially greater time pressure, communication burden, and analytical reconciliation. Long-term sustainability depends on the latter as well as the former.

Operational experience already reflects this distinction. Conjunction alerts frequently trigger bilateral exchanges to determine responsibility, reconcile ephemerides, and confirm maneuver intent. Safe outcomes may still be achieved, yet the process relies on human coordination under narrowing timelines. As encounter frequency rises, success depends increasingly on reaction speed rather than shared expectation. The absence of a systemic indicator prevents regulators and operators from distinguishing between a stable environment and one approaching coordination saturation, understood here as the point at which coordination demands begin to exceed consistently manageable decision timelines.

The proposed OSA would supplement assessment of individual conjunction events by evaluating the predictability and scalability of the broader coordination environment through which collision risk is managed. Stability would describe an orbital regime in which operators can routinely resolve encounters within consistent and sufficient decision margins. In practical terms, this would include maintaining adequate lead time for decision-making, minimizing divergence between independent risk assessments, and limiting cascading maneuver dependencies across multiple operators, where one maneuver triggers additional adjustments by others. Instability would emerge when coordination latency compresses, prediction divergence widens, or maneuver coupling proliferates. The objective is diagnostic: to assess whether operational practices remain scalable as orbital populations grow.

5.2. Participation and organizational structure

Participation would be defined by operational interaction rather than geographic representation. Universal state membership at inception would convert an analytical pilot into a diplomatic negotiation and would likely impede implementation. Instead, actors would participate because their spacecraft already encounter one another within shared orbital regimes.

The initial cohort would be structured around three functional categories. First, civil safety service providers would contribute aggregated indicators derived from warning dissemination and coordination patterns. Second, commercial SSA analytics providers would contribute

indicators reflecting prediction reconciliation and maneuver planning support. Third, spacecraft operators could contribute appropriately anonymized coordination outcomes, including response timing and maneuver execution windows, subject to operational and commercial constraints.

Together, these participants would capture key segments of the conjunction resolution process from detection to action, while acknowledging that participation would remain partial and may expand over time. Participation would be open to any actor meeting the same operational criteria, allowing expansion through demonstrated analytical value rather than political accession.

National authorities, insurance analysts, and technical standards communities would participate as observers. Their role would be interpretive rather than operational: they would assess implications without transforming the pilot into a regulatory process.

Coordination would be performed by a neutral technical secretariat, potentially hosted by an independent, multi-regional research consortium. The secretariat would aggregate, anonymize, and publish results according to a transparent and reproducible methodology. Its mandate would exclude operational direction, compliance evaluation, or policy prescription. Actors would participate by adopting a common reporting protocol rather than acquiring formal institutional membership.

The initiative would operate outside formal intergovernmental frameworks during the pilot phase. Its purpose would be observational rather than regulatory. Results may inform future international discussions, but the pilot itself would function as a pre-institutional analytical mechanism.

5.3. Metrics and construction of the Orbital Stability Index

The proposed OSA would measure operational coordination burden rather than object counts. Raw population size does not determine manageability; coordination demand does. Indicators would therefore be derived from routine conjunction handling across participating systems.

Relevant parameters could include encounter density within shared orbital regimes, lead time between alert and decision point, duration of coordination exchanges, divergence between independent predictions, maneuver coupling frequency, and the occurrence of time-critical resolution conditions. Each parameter would reflect a dimension of operational predictability rather than incident occurrence.

These indicators would be aggregated into the Orbital Stability Index following empirical calibration and validation. The OSI would not assign responsibility or evaluate individual actors. Instead, it would express whether coordination margins remain statistically stable across the system. Increasing urgency or persistent divergence would tend to reduce stability even in the absence of collisions, while consistent resolution windows would indicate more scalable operational practice.

Because data would be anonymized and aggregated, the index would characterize system-level coordination conditions rather than individual operator performance. The output would be informational rather than supervisory. An illustrative example demonstrating how coordination indicators may be translated into an OSI assessment is provided in Appendix A.

Existing literature has developed several approaches for assessing orbital sustainability, environmental capacity, and collision-risk conditions. The Space Sustainability Rating (SSR), for example, is a composite indicator that evaluates missions across multiple dimensions, including debris mitigation, collision avoidance, and data-sharing practices [33]. Environmental- and space-capacity research has similarly sought to characterize the extent to which orbital activity is compatible with a stable evolution of the debris environment and to define, quantify, monitor, and ultimately manage the finite capacity of near-Earth orbital space [33,34]. Related risk-based orbital-classification work further identifies characteristics relevant to assessments of orbital capacity and collision risk, including altitude and orbital persistence, object mass and area-to-mass ratio, maneuver capability and responsiveness, and operational collision-risk thresholds [35]. These approaches address important dimensions of orbital sustainability and operational risk, but OSA and OSI are not intended to duplicate or replace them. Their distinct object of assessment is the system-level coordination environment: OSA evaluates whether the processes through which conjunction risk is assessed and coordinated remain predictable and scalable as operational demands increase, while OSI communicates the resulting system-level assessment. Physical and operational characteristics identified in the existing literature may therefore inform the interpretation and eventual empirical calibration of OSA indicators, including differences in maneuver capability and responsiveness, without converting OSA into an environmental-capacity metric or an operator-rating mechanism.

5.4. Governance significance of a shared diagnostic layer

The proposed OSA would not constitute a traffic management authority. Its function would parallel early safety observation mechanisms that have historically preceded more formal governance arrangements in other domains. Shared measurement would enable actors to recognize common constraints without requiring prior agreement on operational rules.

The index would provide a common evidentiary baseline. Policy discussions could therefore respond to observed patterns of degradation rather than relying solely on hypothetical risk projections. The transition from voluntary coordination to more structured expectations would be informed by observed system behavior rather than precautionary assumptions alone.

Importantly, the assessment would help clarify when decentralized coordination begins to lose scalability. Without a shared indicator, governance proposals may appear anticipatory; with one, they can be grounded in observed conditions. Institutional measures would be more likely to emerge as predictability declines, rather than solely from abstract concerns about risk.

5.5. Role in the progression toward traffic management institutions

The pilot would occupy an intermediate stage between interoperability and governance. Technical harmonization improves encounter resolution; the proposed OSA would evaluate whether those improvements remain sufficient at scale. If stability persists, coordination may remain decentralized. If stability degrades, actors would have a shared evidentiary basis for converging on common operational standards.

In this manner, the assessment would serve as an empirical foundation for subsequent institutional coordination. Standards could acquire justification through observed necessity, and institutional mechanisms may develop in response to demonstrated operational limits. The OSA would therefore neither replace nor predetermine governance structures; rather, it would support their more coherent development.

The following section examines how observed stability conditions could inform the selection of scalable space traffic management architectures.

6. Models of space traffic governance

The analysis above indicates that the orbital environment currently operates through decentralized coordination grounded in professional practice rather than institutional hierarchy, and that technical harmonization and cooperative assessment can improve predictability without redistributing authority. The proposed OSA and its resulting OSI provide a diagnostic basis for evaluating when different governance models may become necessary. The remaining issue therefore concerns governance architecture: if coordination is to scale beyond voluntary convergence, which institutional configurations are structurally compatible with the

characteristics of orbital activity?

Discussions of space traffic management often frame governance as a binary choice between centralized and decentralized systems, a distinction that is analytically incomplete. In orbit, centralization cannot imply operational command analogous to air traffic control: there are no territorial flight regions, no transfer boundaries, and no geographic chokepoints equivalent to ports or national airspace. Orbital motion is continuous and global, and conjunction risk is indifferent to jurisdiction. The relevant distinction is therefore not command versus autonomy, but centralized rulemaking combined with decentralized implementation.

Several governance configurations are particularly relevant in this context: decentralized coordination, federated interoperability networks, centralized institutional standard-setting, and hybrid arrangements combining centralized rulemaking with distributed implementation. These configurations are not mutually exclusive endpoints but analytical models that may emerge sequentially or in combination.

6.1. Decentralized coordination: the existing condition

The current orbital environment is characterized predominantly by decentralized coordination, supported by voluntary information exchange and significant reliance on tracking data derived from military systems [17]. Public SSA services, including EU SST and TraCSS, distribute conjunction data; commercial providers refine analysis; operators negotiate maneuvers directly; and states supervise national actors under Article VI of the Outer Space Treaty [6]. Safety performance depends on a combination of data quality, analytical consistency, and communication practices, supported by established professional norms.

This structure preserves sovereignty, accommodates national security sensitivities, and allows technical innovation. It also reflects prevailing political conditions: states retain operational control of spacecraft and have not demonstrated a clear willingness to transfer real-time decision authority to a supranational body.

However, decentralized coordination does not scale predictably because it lacks shared thresholds, consistent response expectations, and defined coordination timelines. As encounter frequency increases, interpretive divergence and coordination latency increase. Without shared thresholds or structured expectations, coordination becomes more resource-intensive and increasingly time-sensitive. The model remains viable under moderate density but exhibits structural fragility at constellation scale.

Recent policy initiatives reflect a similar pattern of strengthening coordination frameworks. The United States' Space Policy Directive-3 directs the development of civil space traffic management capabilities, focusing on conjunction assessment, data provision, and operator coordination support, while avoiding centralized operational control [36]. The European Union's 2022 approach similarly emphasizes interoperability, information exchange, and the development of shared STM capabilities across Member States [37]. Together, these approaches expand the operational foundation for STM while maintaining distributed implementation.

6.2. Federated interoperability networks

A federated model connects multiple SSA providers through shared technical standards and data exchange protocols [38]. National, regional, and commercial systems operate independently while producing interoperable outputs. Authority remains distributed; consistency derives from harmonization rather than hierarchy.

Elements of this structure are already emerging. EU SST aggregates national contributions, TraCSS is intended to integrate governmental and commercial data, and commercial providers supplement both. The distinction between fragmentation and federation lies in the presence of shared standards. Without harmonization, multiplicity produces conflicting guidance; with harmonization, it can produce redundancy and resilience.

Federation improves operational consistency but does not resolve questions of normative authority. Issues such as acceptable risk thresholds, certification of autonomous collision-avoidance systems, and mandatory coordination expectations cannot be resolved through technical alignment alone. A federated network therefore functions primarily as an infrastructure layer rather than a comprehensive governance regime.

6.3. Centralized institutional standard-setting

Centralization in the orbital context is best understood institutionally rather than operationally. The closest precedent is the International Civil Aviation Organization. ICAO does not control aircraft; instead, it adopts Standards and Recommended Practices (SARPs), which States implement through national regulatory systems [31]. Deviations must be formally notified under Article 38 of the Chicago Convention. Compliance is reinforced through transparency, audit processes, and operational incentives rather than direct supranational enforcement.

An analogous arrangement in space would involve an international body responsible for developing global technical standards, transparency expectations, and review mechanisms, while operational authority remains national. The institution would coordinate the rule framework rather than conduct spacecraft operations.

Such a structure addresses a primary limitation of decentralized coordination: the absence of common thresholds and accountability mechanisms. Internationally recognized standards for conjunction categorization, maneuver notification practices, ephemeris quality, and automated decision-system validation would convert voluntary harmonization into predictable operational expectations.

Importantly, ICAO standards are not adopted by universal consensus but by a governing council exercising delegated authority. This procedural feature allows technical evolution while preserving sovereignty through national implementation and declared deviation.

6.4. Treaty-based consolidation

A more legally consolidated configuration resembles the International Maritime Organization (IMO) framework, which produces binding conventions including the International Convention for the Safety of Life at Sea (SOLAS) and the Convention on the International Regulations for Preventing Collisions at Sea (COLREGs) [32,39]. States implement these obligations through national regulatory systems, while compliance is indirectly reinforced through market access and operational certification mechanisms.

In the space domain, enforcement operates through Earth-based dependencies rather than territorial checkpoints. Launch licensing, reentry authorization, spectrum allocation, ground infrastructure access, insurance underwriting, and financing conditions provide practical implementation pathways. A treaty-based STM framework could link adherence to internationally agreed standards to these regulatory mechanisms.

However, immediate treaty negotiation presents practical difficulties. Technical practices remain in development, and premature codification risks embedding unstable standards. The maritime experience suggests that consolidation typically follows demonstrated operational convergence rather than preceding it.

6.5. Hybrid phased architecture

A hybrid architecture combines elements of the preceding models into a sequential structure. Technical harmonization enables interoperable coordination. Institutional standard-setting then formalizes expectations without requiring binding obligations. Selective treaty-based obligations may follow once operational practice stabilizes and verification mechanisms mature.

This progression mirrors the development of governance in other transportation systems, where shared operational practice preceded standard adoption and legal consolidation followed sustained implementation. The orbital environment presents similar structural constraints: global coupling of risk, absence of territorial boundaries, and reliance on nationally licensed operators.

The governance question is therefore not whether centralization or decentralization will prevail, but how centralized institutional standard-setting and decentralized implementation can be integrated to maintain predictability under increasing activity levels. Decentralized coordination describes the present condition; federated interoperability improves consistency; institutional standard-setting provides scalability; legal consolidation may emerge once supported by stable practice.

The following section applies this framework to outline a phased pathway toward a STMA, aligned with these structural conditions.

7. Toward a Space Traffic Management Authority (STMA)

The preceding analysis indicates that scalable coordination in the orbital environment requires a locus of normative authority distinct from operational control. Technical harmonization alone cannot resolve persistent questions concerning acceptable risk thresholds, coordination expectations, or the validation of automated decision-support systems. At the same time, the physical and legal structure of space activities is incompatible with a centralized traffic management model. The institutional challenge therefore lies between command authority and fragmentation: a coordinating body capable of defining shared expectations while preserving decentralized execution.

Rather than presenting a binary choice between centralized and decentralized governance, this paper advances a phased architecture that integrates interoperability, institutional coordination, distributed enforcement, and selective legal consolidation into a sequential pathway. The objective is not to prescribe a single treaty-based system, but to identify a governance model capable of evolving in alignment with operational practice, technical maturity, and contemporary geopolitical constraints.

7.1. Functional mandate and scope

STMA would not exercise real-time operational control. Its mandate would be normative and supervisory, focused on developing internationally recognized standards for conjunction classification, maneuver coordination expectations, ephemeris quality, data exchange protocols, and validation criteria for automated collision assessment systems [13]. States would incorporate these standards into national licensing and supervisory regimes pursuant to their obligation to authorize and continually supervise national space activities under Article VI of the Outer Space Treaty.

Structurally, STMA would operate through an institutional model analogous to ICAO, comprising a central governing body, specialized technical functions, and a permanent secretariat. A representative assembly or council of participating States would provide strategic direction and adopt standards and recommended practices, while implementation remains anchored in national regulatory systems. Beneath this body, standing technical committees composed of regulators, operators, and subject-matter experts would develop, review, and periodically update operational standards relating to conjunction classification, maneuver coordination, ephemeris quality, data exchange protocols, and system performance. Where operationally necessary, specialist subcommittees could address discrete technical issues such as autonomous collision-avoidance systems, algorithmic validation, or communications interoperability, while remaining integrated within the broader STM framework. A permanent secretariat would support coordination, including convening expert groups, publishing standards and recommended practices, maintaining transparency mechanisms, and facilitating periodic system-level assessments informed by OSA and OSI.

In addition to standard-setting, STMA would support transparency mechanisms, including the publication of recommended practices, notification of deviations, and periodic safety assessments derived from systemic stability monitoring. In this context, OSA and OSI, as proposed in Section 5, provide a shared diagnostic basis for evaluating system-level coordination under evolving operational conditions. These functions enable coordination through shared expectations rather than direct control [40].

STMA would not assume adjudicatory functions concerning liability or fault allocation, consistent with the ICAO model in which responsibility for compliance and enforcement remains with states. Responsibility for damage would continue to be governed by existing treaty frameworks and domestic regulatory regimes. The institutional objective is not centralized command, but the establishment of shared operational expectations, supported by transparency and state-based implementation, enabling consistent and accountable system-level coordination.

7.2. Interoperability as the pre-institutional foundation

Institutional coordination in STM depends on the prior development of shared technical and operational reference layers. In other safety-critical domains, these have emerged before formal institutional consolidation. Global Navigation Satellite Systems (GNSS) provide a globally consistent positioning and timing infrastructure, maintained by multiple sovereign operators yet functioning as a safety-critical reference layer through interoperability and standardization rather than centralized authority [41].

Similarly, multi-stakeholder rulemaking processes such as the Joint Authorities for Rulemaking on Unmanned Systems (JARUS) demonstrate how regulators, industry, and technical experts can converge on common risk classifications, data formats, and operational expectations while preserving national implementation authority [42]. Technical standardization bodies, including the International Organization for Standardization, further support this convergence by establishing harmonized specifications for data exchange, system performance, and interoperability across jurisdictions.

These mechanisms demonstrate that coordination can be structured through shared standards, transparency, and interoperable data frameworks prior to the creation of centralized institutional authority. In the space domain, such alignment is a necessary condition for scalable traffic management, allowing actors to interpret and respond to shared information consistently despite operating within independent legal and operational systems.

7.3. Institutional formation without immediate treaty consolidation

A central institutional question concerns whether the establishment of an STMA requires a binding multilateral treaty comparable to the Chicago Convention. The aviation system is frequently cited as a model, but its institutional formation reflects a particular historical context rather than a necessary legal pathway. The Chicago Convention was negotiated at the end of World War II, in a context that facilitated the creation of a comprehensive treaty-based framework for global civil aviation [31].

The contemporary space environment does not reflect comparable conditions. Space activities are characterized by strategic competition, divergent national interests, and uneven technological capabilities. Under such circumstances, the negotiation of a comprehensive, binding global convention for space traffic management at an early stage is unlikely to be feasible and may risk embedding premature or unstable standards.

Importantly, the functional characteristics associated with ICAO, namely centralized standard-setting combined with decentralized national implementation, do not depend exclusively on treaty origin. In the space domain, a comparable institutional model can emerge without

immediate treaty consolidation. States already exercise regulatory authority over space activities through licensing, authorization, and supervision mechanisms pursuant to Article VI of the Outer Space Treaty. These existing frameworks provide a sufficient legal interface through which internationally developed standards can be incorporated without requiring the prior creation of a binding global convention.

A STMA could therefore be established initially through a process of coordinated institutional formation among participating states, technical bodies, and industry actors, without requiring formal delegation of authority through a binding treaty. Its emergence would be driven by practical convergence: states and operators aligning around common standards, technical bodies contributing to their development, and early adopters incorporating these standards into domestic regulatory frameworks. Over time, sustained participation and consistent use would give rise to a de facto institutional center of standard-setting, with legitimacy derived from technical credibility, adoption, and reliance rather than formal legal mandate.

Recent European Commission discussions have likewise highlighted the potential role of voluntary measures in the development of STM governance. A 2025 DG DEFIS workshop on non-binding measures in support of STM examined the use of standards, certification mechanisms, labelling schemes, and incentive-based approaches as tools for promoting coordination, sustainability, and regulatory convergence in an increasingly congested orbital environment. The workshop further supported a “soft-law-first” approach in which governance mechanisms are tested in practice, evaluated based on demonstrated effectiveness, and progressively formalized over time, reflecting broader interest in governance models that rely upon operational convergence and voluntary coordination mechanisms prior to the establishment of more comprehensive regulatory arrangements [43]. This incremental approach is broadly consistent with the implementation logic reflected in the IAA's 2018 STM roadmap, which similarly recognized that effective STM governance is likely to emerge through the progressive convergence of technical, operational, regulatory, and institutional measures rather than through the immediate creation of a comprehensive global regime [3].

Only at a later stage, once operational practices stabilize and convergence is demonstrated, would selective treaty-based obligations become appropriate. In this respect, the trajectory more closely reflects the development of the IMO framework, in which binding conventions largely codified established practices rather than initiating them [44].

7.4. Institutional analogy: ICAO-style coordination

The closest structural precedent for institutional standard-setting is the International Civil Aviation Organization, which does not exercise operational control over aircraft but instead adopts Standards and Recommended Practices (SARPs) that States implement through national regulatory systems [31]. Deviations are formally notified under Article 38 of the Chicago Convention, enabling transparency without requiring uniform compliance.

Importantly, ICAO technical standard-setting does not require unanimity among all States. These standards are adopted through the ICAO Council and implemented through national regulatory systems, allowing technical requirements to evolve without amending the underlying treaty [32,44]. This procedural structure enables timely adaptation in a dynamic operational environment while preserving state sovereignty through national implementation and transparent deviation practices.

A STMA operating on a similar model would convene technical panels composed of regulators, operators, and subject-matter experts, issuing standards and recommended practices addressing conjunction coordination, data exchange, ephemeris quality, and automated decision-support systems. States would integrate these standards into licensing, supervision, and operational authorization processes.

The relevance of this model lies in its procedural structure: centralized rule development can coexist with state sovereignty when implementation remains national, and deviations are transparent. For space operations, where enforcement is mediated through Earth-based regulatory interfaces rather than through in-orbit control, such a model is structurally compatible.

7.5. Phased governance architecture and hybrid enforcement

The governance pathway proposed in this paper is explicitly sequential. In the near term, STM could adopt an ICAO-style structure of centralized standard-setting combined with decentralized implementation, supported by shared operational reference layers and transparency mechanisms. This stage enables consistent interpretation of operational risk without requiring supranational control.

As coordination demands increase, the system could transition into a hybrid compliance phase. In this stage, standards remain centrally defined, but compliance is reinforced through distributed mechanisms embedded in existing regulatory and market structures, including launch licensing, insurance underwriting, spectrum allocation, and access to ground infrastructure. Bilateral and multilateral arrangements, along with technical standardization processes, support this phase without requiring direct supranational enforcement.

At a later stage, once operational practice has stabilized and verification mechanisms are sufficiently mature, selective treaty-based obligations may be introduced. This stage resembles an IMO-style consolidation, limited to specific domains such as debris mitigation performance or maneuver notification, where binding commitments become both technically feasible and politically sustainable.

The resulting architecture is neither fully centralized nor purely decentralized but layered. Centralized rule development, federated operational implementation, and distributed enforcement are deliberately sequenced. Full legal consolidation is not a prerequisite for coordination, but a potential outcome of sustained interoperability, institutional convergence, and demonstrated operational necessity.

8. Implementation challenges and institutional viability

The preceding sections have outlined a phased pathway from enhanced SSA toward progressively institutionalized forms of STM. While this framework seeks to balance technical feasibility with political realism, its long-term viability depends upon addressing a number of practical challenges associated with implementation. In particular, questions concerning compliance, participation, assessment integrity, and geopolitical fragmentation become increasingly important as orbital activity expands and operational environments become more complex. The following sections therefore examine several challenges likely to influence the development of STM governance, including the role of insurance-based incentives, participation by operators possessing advanced SSA capabilities, the integrity of assessment frameworks under conditions of partial participation, and the implications of geopolitical fragmentation for institutional resilience.

8.1. Insurance-based incentive structures and compliance verification

A central challenge for any STM framework concerns not the development of standards themselves, but the mechanisms through which those standards acquire practical influence. The preceding discussion has proposed a model of centralized standard-setting combined with decentralized implementation; however, standards alone do not ensure compliance. The relevant question is therefore not whether STMA would possess direct enforcement authority, but how adherence to its standards could become consequential within the broader operational environment. If compliance remains entirely voluntary and disconnected from operational incentives, STMA risks becoming another international body whose recommendations are acknowledged but routinely disregarded.

One potential pathway lies in the incorporation of STM standards

into insurance underwriting and risk-assessment processes. The maritime sector provides a particularly relevant precedent because influence is exercised through a layered governance structure rather than direct centralized enforcement. International organizations establish technical and operational standards, classification societies verify compliance with those standards, and insurers incorporate those assessments into underwriting decisions, coverage conditions, and renewal determinations [45]. Compliance therefore becomes operationally significant not because a central authority directly compels it, but because multiple institutions rely upon a common set of recognized technical expectations. A comparable structure could emerge in the orbital domain. STMA would function primarily as a standard-setting body; independent auditing, certification, or verification entities would assess compliance against recognized STM standards; and insurers would incorporate those assessments into operational risk evaluations.

Under such an arrangement, STMA would not certify operators directly, nor would insurers independently evaluate conjunction practices, maneuver coordination procedures, or ephemeris quality. Instead, compliance could be assessed through recognized verification mechanisms operating against internationally recognized STM standards. These assessments would provide an evidentiary basis upon which insurers evaluate operational risk and determine underwriting conditions. Verification could encompass adherence to maneuver-notification procedures, maintenance of minimum ephemeris quality requirements, responsiveness during high-risk conjunction events, participation in recognized coordination frameworks, and compliance with transparency expectations relating to conjunction management. The objective would not be to determine whether an operator has achieved perfect safety performance; rather, it would be to evaluate whether its operational practices are consistent with standards designed to reduce systemic risk and improve predictability across a shared orbital environment.

Such assessments could then inform differentiated underwriting structures. Operators demonstrating sustained compliance with recognized STM standards could be classified as lower-risk policyholders and receive more favorable premium schedules, lower deductibles, broader coverage terms, or preferential renewal conditions. Conversely, repeated failures to respond to conjunction notifications, persistent noncompliance with established coordination procedures, refusal to participate in recognized coordination frameworks, or documented patterns of behavior inconsistent with recognized STM standards could result in premium surcharges, increased deductibles, restrictions on coverage, or adverse renewal determinations. Coverage consequences would therefore not arise from isolated incidents alone, but from demonstrated patterns of conduct indicative of elevated operational risk. In this respect, compliance becomes economically relevant because predictable coordination practices reduce uncertainty not only for operators, but also for insurers assessing potential liability exposure.

8.2. Participation incentives in an era of advanced SSA capabilities

Beyond compliance verification, a second challenge concerns participation incentives for operators possessing increasingly sophisticated internal SSA capabilities. Large constellation operators may possess limited need for externally provided conjunction services, raising questions regarding the long-term relevance of cooperative governance frameworks. SpaceX's February 2026 announcement of the Stargaze platform illustrates this possibility [19]. If proprietary systems continue to mature, some operators may become capable of conducting conjunction assessment, maneuver planning, and operational traffic-management functions at a level comparable to, or in some respects exceeding, that of many public systems. Under such circumstances, the relevance of STMA cannot depend upon providing superior technical services; any framework built upon that assumption would remain vulnerable as private capabilities continue to evolve.

The continued relevance of STMA therefore does not depend upon providing the most accurate conjunction assessments or the most sophisticated analytical capabilities. Rather, its value derives from enabling coordination among actors operating different systems, using different data sources, and functioning under different regulatory and operational constraints. Indeed, proprietary systems such as Stargaze may ultimately function not as alternatives to STM governance, but as participants within a broader coordination architecture operating through shared standards and interoperability mechanisms. Even operators possessing advanced internal capabilities must continue to coordinate with external actors operating under different analytical models, regulatory frameworks, and operational constraints. The governance challenge is therefore not conjunction management within a single constellation, but coordination across a heterogeneous orbital environment composed of states, commercial operators, civil safety services, and private SSA providers. In this context, STMA standards function less as a substitute for proprietary systems than as an interoperability framework through which diverse systems can coordinate activities within a shared orbital environment.

8.3. Assessment integrity and the free-rider challenge

The existence of advanced proprietary capabilities also raises questions regarding the integrity of cooperative assessment frameworks. A dominant operator may be capable of benefiting from improvements in system-wide coordination while contributing limited information to collective assessment mechanisms. If enough major actors adopt this position, the value of shared diagnostic frameworks and common operational standards may erode. This challenge is particularly relevant to the OSA framework and its resulting OSI, both of which depend upon participation by major operators to provide an accurate assessment of coordination conditions across the orbital environment. If dominant actors systematically withhold coordination timing data or other operational indicators, OSI may increasingly reflect the behavior of participating operators while underrepresenting those actors exerting the greatest influence on overall orbital dynamics. Participation therefore becomes important not only for the accuracy of the assessment itself, but also for the credibility of any governance framework built upon it.

One potential means of addressing this concern is to integrate OSA-derived assessments into broader verification and risk-evaluation processes. Certification and auditing mechanisms may separately establish whether individual operators comply with recognized STM standards; OSA would perform a different function by providing insurers, regulators, procurement authorities, and other stakeholders with a longitudinal assessment of system-level coordination conditions across time. Participation in recognized reporting and assessment frameworks could itself provide evidence of transparency and willingness to contribute to collective risk understanding, without converting OSA or OSI into an assessment of individual operator performance. Persistent refusal to contribute relevant operational data may nevertheless increase uncertainty in external risk-evaluation processes. Because insurers routinely distinguish between demonstrable risk management and the absence of verifiable information, operators may possess incentives to participate in OSA processes even where they derive limited direct technical benefit from the resulting assessments.

Insurance-based incentives provide one means of reinforcing this dynamic, but they are unlikely to operate in isolation. As STM standards become progressively incorporated into procurement requirements, financing decisions, launch licensing frameworks, spectrum-access considerations, and other mechanisms governing participation in orbital activities, adherence may acquire significance beyond underwriting alone. Large operators may possess the resources to absorb higher premiums if they choose not to participate in recognized assessment and reporting frameworks; however, the cumulative effect of multiple institutions relying upon common standards may gradually increase the operational and commercial costs associated with nonparticipation. Under such conditions, participation becomes valuable

not because STMA compels it, but because insurers, regulators, procurement authorities, launch providers, and other stakeholders increasingly rely upon shared standards and independently verifiable information concerning coordination conditions and risk-management practices when evaluating operational risk. The influence of STMA therefore derives not from any single enforcement mechanism, but from the progressive integration of its standards into the broader ecosystem through which orbital activities are authorized, financed, insured, and conducted.

8.4. Governance under conditions of partial participation

The phased pathway proposed throughout this paper envisions progressively greater structure in STM governance, beginning with diagnostic assessment through OSA and its resulting OSI and potentially developing toward more formalized standard-setting through STMA. While such a trajectory remains possible, it should not be treated as the sole measure of institutional success. A more demanding test concerns whether the framework remains viable under conditions of geopolitical fragmentation and partial participation. The relevant question is therefore not whether all major spacefaring actors ultimately adopt a common governance architecture, but whether meaningful coordination remains possible if they do not.

The possibility of fragmentation warrants serious consideration. Space activities increasingly involve capabilities that extend beyond technical safety considerations and intersect with national security, economic competitiveness, industrial policy, and strategic autonomy. Under such circumstances, states may have incentives to preserve independent decision-making authority over infrastructure, data, and operational capabilities that they regard as strategically significant. China's non-membership in the Missile Technology Control Regime (MTCR), despite its stated adherence to many MTCR control principles, illustrates that strategic technology governance may develop through partial alignment rather than formal institutional integration [46]. Similarly, restrictions associated with the International Traffic in Arms Regulations (ITAR) contributed to the emergence of alternative supply chains, indigenous technological capabilities, and efforts to reduce dependence upon externally controlled technological ecosystems [47,48]. The relevance of these examples lies not in export controls themselves, but in the broader tendency of states to maintain independent governance authority over capabilities perceived as strategically consequential. STM may prove susceptible to similar dynamics because SSA data, conjunction assessment, maneuver coordination, and traffic-management functions increasingly possess significance extending beyond purely technical safety considerations.

Many of these characteristics are already visible within the contemporary orbital environment. China's development of the Guowang constellation through China SatNet reflects a state-backed effort to establish a large-scale LEO communications infrastructure, with current plans envisioning nearly 13,000 satellites [49,50]. A constellation of such scale creates strong incentives to maintain national capabilities for conjunction assessment, maneuver planning, and traffic-management decision-making. The significance of this development is not merely that China may build a parallel STM architecture, but that reliance upon an externally governed traffic-management framework for functions integral to national space infrastructure may be viewed as strategically undesirable. Similar considerations may influence Russia, emerging space powers, regional coalitions, or any actor seeking to preserve operational autonomy within an increasingly contested orbital environment. Under such circumstances, institutional fragmentation should be viewed not as an exceptional outcome, but as a plausible baseline scenario that any durable STM framework must be capable of accommodating. The more significant question therefore becomes not whether multiple governance architectures emerge, but how coordination may be maintained among them once they do.

8.5. Alternative pathways to institutional development

If fragmentation is treated as a plausible baseline scenario, the relevant question becomes not whether institutional diversity emerges, but how STM governance evolves under such conditions. Under these circumstances, at least three broad institutional pathways may emerge.

The first involves progressive convergence around common governance frameworks, with OSA, OSI, and STMA attracting increasingly broad participation over time. This scenario is broadly consistent with the phased pathway described throughout this paper and represents the most comprehensive realization of the proposed framework.

A second possibility involves functional interoperability without full institutional convergence. Under such an arrangement, states and regional groupings maintain independent STM institutions, assessment methodologies, and governance structures while agreeing upon common operational principles relating to conjunction notification, ephemeris exchange, emergency coordination procedures, and collision avoidance practices. In this scenario, OSA would continue to provide a diagnostic assessment mechanism for participating actors, with OSI communicating the resulting system-level assessment, while STMA functions as a venue for standards development, technical coordination, and procedural harmonization among otherwise distinct governance systems. Institutional diversity persists; however, operational coordination remains possible through shared technical expectations and agreed interface mechanisms.

A third possibility involves the emergence of parallel governance architectures. One plausible manifestation of this scenario would involve a STMA framework initially adopted by participating actors, including the United States, European partners, Japan, India, and other willing states, while other actors maintain independent STM institutions linked to domestic SSA capabilities, constellation-management systems, and national regulatory frameworks. Such an outcome should not necessarily be interpreted as institutional failure. Rather, it would reflect a governance environment in which multiple architectures coexist while attempting to coordinate activities within a shared orbital domain. Under these circumstances, STMA would function less as a universal traffic-management authority and more as a standards-development and coordination institution for participating actors. Its practical role would include the development of interface protocols, information-exchange mechanisms, conjunction-notification procedures, emergency coordination practices, and interoperability standards capable of facilitating predictable interactions between participating and non-participating governance systems. The objective would therefore not be to eliminate governance diversity, but to reduce the operational risks associated with it.

This distinction has important implications for the role of each component of the framework. Under conditions of fragmentation, OSA would continue to provide an independent methodology for assessing coordination conditions within participating communities, while the resulting OSI would synthesize observable indicators into a system-level assessment of coordination stability where sufficient information is available. Assessment coverage may become uneven; however, participating operators, insurers, regulators, procurement authorities, and other stakeholders may still derive significant value from transparent and independently verifiable assessments of system-level coordination conditions. Similarly, STMA would continue to provide a forum for standards development, technical coordination, information exchange, and procedural harmonization among participating actors even if universal adoption remains unattainable. Under such circumstances, STMA's role would not necessarily be to replace national or regional STM architectures, but rather to facilitate the development of common coordination expectations, interface standards, and interoperability mechanisms capable of supporting predictable interactions among diverse governance systems. Participation in such arrangements need not require agreement on broader governance questions. Rather, the practical necessity of coordinating activities within a shared orbital

environment may create incentives for the development of common procedures and interface mechanisms even where institutional diversity persists.

In practice, the resulting governance landscape may resemble aspects of contemporary Internet governance more closely than traditional aviation governance. Internet governance has historically developed through distributed, multi-stakeholder arrangements involving governments, private actors, technical bodies, and international organizations rather than through a single centralized authority [51]. That model is imperfect for STM, which involves safety-critical physical operations rather than digital coordination alone; however, it illustrates that interoperability can be maintained despite institutional diversity. Importantly, the existence of multiple governance architectures does not diminish the value of coordinating institutions. Rather, fragmentation increases the importance of venues capable of facilitating technical harmonization, developing interface standards, supporting information exchange, and reducing operational uncertainty among actors operating under different governance frameworks. Under such circumstances, STMA's relevance derives not from exercising universal authority, but from facilitating the development of common coordination expectations capable of supporting predictable interactions among diverse governance architectures operating within a shared orbital environment.

The viability of the proposed framework should therefore be evaluated not according to its ability to eliminate governance diversity, but according to its ability to facilitate coordination across diverse governance architectures. If OSA, OSI, and STMA require universal participation to remain effective, the framework is unlikely to succeed. If, however, OSA and OSI can provide meaningful diagnostic information under conditions of partial participation, while STMA can support predictable coordination and structured relationships across participating and non-participating governance architectures, fragmentation becomes a design constraint rather than a fatal flaw. Multiple governance architectures may coexist; the more significant challenge concerns the development of mechanisms capable of supporting predictable interactions among actors operating under different procedural assumptions, coordination frameworks, and traffic-management systems. Under this approach, the long-term value of STMA derives not from replacing alternative governance architectures, but from providing an institutional mechanism through which coordination, standardization, and interoperability can progressively develop despite their continued existence. Fragmentation does not necessarily preclude institutional development; rather, it may shape the pathway through which such development occurs.

8.6. Interoperability across governance architectures

The existence of multiple STM governance architectures raises a practical question: how should coordination occur when operators functioning under different institutional frameworks encounter one another during conjunction events? This issue becomes particularly important under conditions of fragmentation, where operators may rely upon different assessment methodologies, procedural assumptions, coordination mechanisms, or regulatory requirements. A collision-risk event involving a satellite operating under a STMA-aligned framework and a satellite operating under an independent governance architecture illustrates the challenge. In such circumstances, institutional resilience depends not upon universal agreement regarding governance structures, but upon the existence of mechanisms capable of supporting predictable operational interactions across governance boundaries.

The immediate objective of STM governance should therefore not necessarily be the establishment of universally accepted right-of-way rules. While greater harmonization regarding maneuver responsibility may emerge over time, achieving comprehensive agreement at an early stage would likely encounter many of the same geopolitical obstacles discussed in the preceding sections. A more realistic near-term objective involves the development of procedural interoperability capable of supporting coordination among operators functioning under different governance architectures. Operators may disagree regarding governance structures, regulatory approaches, or broader institutional arrangements while still agreeing upon common notification procedures, communication protocols, emergency coordination mechanisms, and minimum information-sharing requirements. Under this approach, coordination becomes possible even in the absence of complete institutional convergence.

Such interoperability mechanisms could be developed through STMA and may include standardized conjunction-notification formats, common timelines for operator responses, minimum information-sharing expectations, agreed escalation procedures for high-risk events, and designated communication channels for urgent coordination. These mechanisms would not require operators to adopt identical governance frameworks. Rather, they would establish a common operational language through which different systems can interact during conjunction events. In practice, the objective would be to reduce uncertainty regarding how another operator is likely to communicate, respond, and coordinate under time-sensitive conditions.

The operational importance of procedural interoperability becomes apparent when considering a conjunction involving operators functioning under different governance architectures. One system may assign maneuver responsibility based upon spacecraft maneuverability, another may prioritize mission criticality, while a third may rely upon different procedural assumptions altogether. In the absence of interoperability mechanisms, each operator could reasonably expect the other to maneuver, creating uncertainty precisely when decision timelines become increasingly constrained.

Under a STMA-supported interoperability framework, however, the immediate objective would not be to resolve every disagreement regarding maneuver responsibility. Instead, operators would be expected to follow common coordination procedures, including standardized conjunction notifications, acknowledgement requirements, communication timelines, escalation protocols, and confirmation of maneuver intent. If an operator does not intend to maneuver, is unable to maneuver, or fails to respond, that information would be communicated through predefined channels, allowing the remaining operator to make informed decisions based upon a shared procedural framework rather than assumptions regarding the behavior of another governance system. While procedural interoperability cannot guarantee agreement regarding maneuver responsibility, it can ensure that disagreement is identified, communicated, and managed through established coordination processes rather than emerging unexpectedly during time-critical decision windows.

In this respect, procedural interoperability reduces the risk that uncertainty regarding responsibility becomes a source of operational instability. More broadly, it illustrates how STMA could support coordination among diverse governance architectures without requiring immediate agreement on all aspects of STM governance. The objective is not to eliminate institutional diversity, but to ensure that operators functioning under different governance systems can nevertheless interact through predictable procedures, shared expectations, and established communication mechanisms when orbital safety requires coordination.

8.7. Right-of-way and institutional resilience

The development of right-of-way principles presents a related but distinct challenge. Current STM practices largely rely upon voluntary coordination, operator judgment, and ad hoc communication rather than universally accepted priority rules. As orbital congestion increases, however, pressure may emerge for greater procedural predictability. Questions concerning whether maneuvering responsibility should depend upon spacecraft maneuverability, mission type, operational status, orbital regime, or other factors remain unresolved. As orbital activity increases, the development of more predictable approaches to

maneuver responsibility may become increasingly important for reducing uncertainty during conjunction events involving operators functioning under different governance architectures.

The initial governance challenge is therefore not the immediate creation of universally accepted right-of-way rules, but the establishment of interoperable coordination mechanisms capable of functioning despite the absence of such rules. Once operators are capable of exchanging information, communicating intentions, and coordinating responses through common procedures, more complex questions concerning maneuver responsibility become more manageable. In this respect, interoperability may be viewed as a prerequisite for meaningful discussions concerning right-of-way rather than a substitute for them.

Importantly, the absence of universally accepted right-of-way rules does not imply the absence of coordination. Maritime governance developed operational procedures long before achieving comprehensive international harmonization, and civil aviation similarly evolved through incremental procedural standardization rather than immediate agreement on all substantive issues. STM may follow a comparable pathway. The initial objective should therefore be the development of interoperable coordination procedures capable of functioning across different governance architectures while allowing more complex questions concerning priority rules and maneuver responsibilities to mature over time through operational practice, technical experience, and continued international dialogue.

Under this approach, STMA would not function as a centralized authority determining the outcome of individual conjunction events. Rather, its role would be to facilitate the development of procedural standards, coordination practices, technical guidance, and eventually recommendations concerning maneuver responsibility as operational experience accumulates. The long-term significance of such an institution therefore lies not in its ability to impose immediate solutions to unresolved governance questions, but in its capacity to provide a structured environment through which those questions can be examined, tested, and progressively refined over time.

9. Conclusion

SSA remains indispensable but incomplete: public and commercial systems routinely track objects, generate conjunction assessments, and support coordination among operators, yet awareness is still constrained by uneven tracking coverage, asymmetries in data quality, dependence in many cases on tracking data derived from military systems, and the voluntary nature of key forms of operator disclosure. What does not yet exist is a coherent STM regime in the institutional sense. Collision avoidance occurs daily, but it remains negotiated rather than structured, and predictability still depends too heavily on bilateral communication, professional practice, and converging assumptions rather than on shared governance architecture.

The central challenge is therefore not simply to expand awareness, but to convert awareness and coordination into scalable management. SSA provides information about orbital conditions; STM requires agreed procedures, shared operational expectations, and institutions capable of stabilizing how actors respond to that information. Without such common baselines, increasing traffic density turns coordination into a progressively fragile process in which negotiation burden, interpretive divergence, and time pressure accumulate faster than existing practices can reliably absorb them.

This paper advances a phased institutional pathway for that transition. The first step is pre-institutional alignment. Short-term operational improvements can establish interpretive interoperability without redistributing authority: standardized risk categories, harmonized maneuver communication windows, minimum ephemeris quality conventions, and common terminology across safety services. These measures are supported conceptually by models of interoperable reference-layer coordination in other safety-critical domains. Global Navigation Satellite Systems (GNSS) show how globally trusted, safety-critical information can be coordinated across sovereign actors through interoperability rather than centralized command. Multi-stakeholder processes such as the Joint Authorities for Rulemaking on Unmanned Systems (JARUS), together with technical standardization bodies such as the International Organization for Standardization (ISO), illustrate how common data formats, risk classifications, and operational expectations can converge before formal institutional consolidation. In the orbital domain, that kind of alignment is not the end state; it is the necessary starting point.

The second step is diagnostic. This paper proposes a cooperative Orbital Stability Assessment (OSA) as a pilot framework for evaluating system-level coordination conditions, producing a periodic Orbital Stability Index (OSI) that synthesizes relevant indicators into an assessment of whether coordination remains predictable and scalable as orbital activity increases. The proposed pilot would not function as a regulatory body; it would bring together a limited operational cohort including civil safety service providers, commercial SSA analytics providers, and spacecraft operators, with national authorities, insurance analysts, and technical standards communities participating in an observational capacity. Its purpose would be to generate shared empirical understanding of systemic conditions by measuring coordination burden rather than incident occurrence. In that way, the OSA and OSI would give regulators, operators, and other stakeholders a common evidentiary basis for determining when decentralized coordination remains sufficient and when it begins to lose scalability.

The third step is institutional. Building on that empirical foundation, this paper proposes a Space Traffic Management Authority (STMA) as a centralized standard-setting institution that would not exercise real-time operational control. Its function is best understood through the institutional logic of the International Civil Aviation Organization (ICAO): centralized rule development combined with decentralized national implementation. STMA would define internationally recognized operational expectations while implementation would remain anchored in national licensing and supervisory systems. That structure is particularly important because the relevant question is not whether STM should be centralized or decentralized, but how centralized institutional standard-setting and decentralized implementation can be integrated to preserve predictability under increasing orbital density.

Importantly, such a structure does not require the prior creation of a comprehensive treaty equivalent to the Chicago Convention, which established ICAO in a very different historical and political context. The contemporary space environment is marked by strategic competition, divergent national interests, and uneven technological capabilities; in that political environment, immediate treaty-based institutional creation is unlikely to be realistic and may entrench unstable standards prematurely. What is functionally significant in the ICAO model is not its treaty origin alone, but its procedural structure: ICAO technical standard-setting does not depend on unanimity among all States; it operates through institutional procedures centered on the ICAO Council, allowing standards to evolve without continual treaty amendment. In present conditions, that distinction matters. A model that depends on universal consensus is less likely to adapt quickly enough than one that permits institutional standard-setting while preserving sovereignty through national implementation and transparent deviation.

As coordination demands intensify, the pathway proposed here then moves into a hybrid compliance phase. Standards remain centrally defined, but compliance is reinforced through distributed mechanisms already embedded in terrestrial regulatory and market systems, including launch licensing, spectrum allocation, insurance, financing, and access to ground infrastructure. Only at a later stage, once operational practice has stabilized and verification mechanisms are sufficiently mature, could selective treaty-based obligations become viable in narrow fields such as debris mitigation performance or maneuver notification. In that respect, the longer-term trajectory is closer to the development of the International Maritime Organization (IMO), in which binding conventions largely codified established practices rather

than initiating them.

The proposed architecture is therefore a deliberately phased system: pre-institutional interoperability, pilot-based systemic assessment, centralized institutional standard-setting, distributed compliance reinforcement, and only then, where politically and technically sustainable, selective legal consolidation. STMA is not conceived as a global traffic controller; it is conceived as an institutional locus for coordinated rulemaking within a framework of federated operations and state-based implementation.

Importantly, the viability of this pathway does not depend upon universal participation. The analysis in Section 8 demonstrates that STM governance may develop under conditions of partial participation, institutional diversity, asymmetric capabilities, and geopolitical fragmentation. The framework therefore does not assume that all major operators, states, or regional groupings ultimately converge within a single governance architecture. Indeed, large constellation operators may increasingly possess sophisticated proprietary SSA and coordination capabilities, potentially reducing their reliance upon externally provided conjunction services. Under such circumstances, the value of STMA derives not from providing superior technical capabilities, but from enabling coordination among actors operating different systems, using different data sources, and functioning under different regulatory and operational constraints.

The proposed framework likewise does not depend upon direct supranational enforcement. As operational practices mature, compliance may be reinforced through distributed mechanisms already embedded within terrestrial regulatory and market structures, including insurance underwriting, launch licensing, spectrum allocation, procurement requirements, financing conditions, and access to critical infrastructure. Participation therefore becomes valuable not only because of the direct benefits of coordination, but also because multiple institutions may progressively incorporate STM standards into risk assessment, authorization, and commercial decision-making processes. In this respect, adherence may acquire practical significance even in the absence of centralized enforcement authority.

Finally, fragmentation should not necessarily be interpreted as institutional failure. Multiple governance architectures may emerge, coexist, and evolve simultaneously. Under such conditions, the primary function of STMA is not to eliminate governance diversity or replace alternative frameworks, but to facilitate predictable interaction among them through common standards, interoperability mechanisms, and shared procedural expectations. The long-term resilience of STM governance will therefore depend not only upon institutional design, but also upon its ability to remain operationally relevant across diverse technical, commercial, regulatory, and geopolitical environments.

STM will not emerge through a single legal instrument or organizational decree. It will emerge through cumulative convergence: first in shared interpretation of conjunction information, then in common assessment of systemic stability, then in centralized institutional standard-setting linked to decentralized implementation, and only later, where viable, in selective legal consolidation. The transition from SSA to STM is therefore not merely technological. It is institutional, procedural, and sequential [40]. The task is to build that transition deliberately, before operational complexity forces it reactively.

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Appendix A. Illustrative OSA-to-OSI Assessment Example

This appendix provides an illustrative example of how Orbital Stability Assessment (OSA) could evaluate coordination conditions within an orbital regime and translate those observations into an Orbital Stability Index (OSI) assessment. The values presented below are hypothetical and are intended solely to demonstrate the proposed methodology. They should not be interpreted as validated operational thresholds, recommended performance targets, or empirically calibrated benchmarks.

The illustrative parameters used in this example—including assessment periods, response timelines, normalization approaches, and maneuver-coupling observation windows—have been selected for explanatory purposes. Actual implementation would require empirical calibration using historical conjunction data, operator practices, SSA-provider inputs, and operational validation across different orbital regimes. Accordingly, the numerical values below are not intended to represent typical operational conditions within any specific orbital shell, constellation, operator population, or operational dataset. Rather, they have been selected to demonstrate how differing coordination environments may produce different stability assessments even where the realized collision outcome is held constant for illustrative purposes.

Assessment Period (illustrative): 30 days.

Normalization Assumption (illustrative): Where applicable, values are normalized by the number of active spacecraft in the assessed orbital regime to facilitate comparison between regimes of different sizes. Alternative normalization methodologies may be appropriate depending upon implementation context.

Realized Collision Outcome (illustrative): No collision occurs in either shell during the assessment period. This assumption is used solely to hold the realized collision outcome constant for purposes of comparing coordination conditions and does not imply equivalent conjunction risk, aggregate collision risk, or risk reduction across the two shells.

A.1. Illustrative Indicator Definitions

Indicator	Illustrative Measurement
Encounter Density	Coordination-relevant conjunction events per active spacecraft during the assessment period
Prediction Divergence	Percentage of coordination-relevant events for which participating SSA providers produce materially different conjunction assessments, such as differing risk classifications, miss-distance estimates, or maneuver recommendations
Coordination Latency	Median time between issuance of a coordination-relevant alert and confirmation of operator maneuver intent or non-maneuver decision
Maneuver Activity	Avoidance maneuvers per active spacecraft during the assessment period
Maneuver Coupling	Secondary coordination events generated by avoidance maneuvers within a defined observation window (illustratively assumed here to be 72 h)

Where prediction divergence is observed, OSA would, where practicable, record the identified source or contributing factors underlying the divergence rather than treating the divergence rate alone as dispositive.

A.2. Illustrative Shell A: Lower Coordination Burden

Indicator	Value
Active Spacecraft	100
Coordination-Relevant Conjunction Events	40
Encounter Density	0.40 (40/100)
Prediction-Divergence Rate	7.5%
Median Coordination Latency	6 h
Avoidance Maneuvers	4
Maneuver Activity	0.04 (4/100)
Secondary Coordination Events Following Maneuvers	1
Secondary Coordination Events per Avoidance Maneuver	0.25 (1/4)

A.3. Illustrative Shell B: Higher Coordination Burden

Indicator	Value
Active Spacecraft	100
Coordination-Relevant Conjunction Events	40
Encounter Density	0.40 (40/100)
Prediction-Divergence Rate	30%
Median Coordination Latency	18 h
Avoidance Maneuvers	11
Maneuver Activity	0.11 (11/100)
Secondary Coordination Events Following Maneuvers	18
Secondary Coordination Events per Avoidance Maneuver	1.64 (18/11)

A.4 Illustrative Maneuver-Coupling Event

The concept of maneuver coupling is intended to capture situations in which one avoidance maneuver generates additional coordination requirements for other actors.

Illustratively:

Event Sequence	Coordination Effect
Initial conjunction alert	Operator A evaluates a conjunction event and determines that a maneuver is necessary
Avoidance maneuver by Operator A	The trajectory change generates three additional coordination-relevant conjunction assessments involving nearby spacecraft
Secondary coordination activity	Two neighboring operators exchange coordination information and reassess conjunction risk
Follow-on maneuver	One neighboring operator performs an additional avoidance maneuver
Result	The original event generates additional coordination burden extending beyond the initially affected spacecraft

Under the proposed framework, this event would be characterized as a coupled maneuver environment because the original avoidance action produced additional coordination requirements elsewhere in the orbital regime. The significance is not that a collision occurred; none did. Rather, the event generated additional coordination effort and greater operational interdependence among participating actors.

A.5 Illustrative Interpretation

Although Illustrative Shell A and Illustrative Shell B have the same realized collision outcome during the assessment period, they differ materially in coordination burden.

OSA is intended to supplement, rather than replace, established conjunction- and collision-risk assessment by evaluating an additional dimension: the coordination burden associated with managing conjunctions across an orbital regime.

Indicator	Shell A	Shell B	Illustrative Interpretation
Encounter Density	0.40	0.40	Similar conjunction volume
Prediction-Divergence Rate	7.5%	30%	Greater analytical disagreement in Shell B
Median Coordination Latency	6 h	18 h	Longer coordination time in Shell B
Maneuver Activity	0.04	0.11	Greater maneuver activity in Shell B
Secondary Coordination Events per Avoidance Maneuver	0.25	1.64	Greater coordination interdependence in Shell B

Under the proposed framework, the indicator set would ultimately be synthesized into an overall OSI assessment. This illustrative example does not assign a numerical OSI score because the weighting, normalization, and aggregation of individual indicators would require empirical calibration and

validation across operational datasets and orbital regimes. The indicator profile for Illustrative Shell B nevertheless reflects greater coordination burden than that of Illustrative Shell A, including more frequent analytical disagreement, longer coordination timelines, greater maneuver activity, and more secondary coordination activity following avoidance maneuvers.

The purpose of this example is not to establish operational standards or final OSI thresholds. Rather, it demonstrates how OSA could distinguish between orbital environments exhibiting materially different levels of coordination burden even where the realized collision outcome is held constant for illustrative purposes. In this manner, OSA evaluates whether coordination processes supporting collision avoidance remain predictable and scalable as orbital activity, coordination demands, and operational interdependence increase, while OSI communicates the resulting system-level assessment.

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